

Session 6 Quiz - Training Your Own AI Model

Track B | Grades 9–12 | 20 Questions

Q1 [Type: MCQ | Difficulty: Easy | QID : 52742]

When training, you compare the model's predictions to the correct answers and measure the difference; this difference is called the _____ (or error).

- A. Logo
- B. Cable
- C. Loss
- D. Memory

Correct Answer: C

Explanation: "Loss" measures how wrong the model's predictions are. Training works by minimising this loss - adjusting parameters so the model's predictions get closer to the correct answers over time. Memory, logo, and cable have no role in this calculation.

Q2 [Type: MCQ | Difficulty: Easy | QID : 52743]

When you train a model and its performance on training data keeps improving but its performance on new data starts getting worse, this is a sign of _____.

- A. Overfitting
- B. Compression
- C. Translation
- D. Logo

Correct Answer: A

Explanation: Overfitting is when the model learns the specific examples in training data rather than generalisable patterns. The diverging accuracy curves (training high, validation low) is the classic overfitting signature - and exactly what you see if you train for too many epochs on too little data

Q3 [Type: MCQ | Difficulty: Easy | QID : 52744]

In tools like Google Teachable Machine, one complete pass through all the training images is called an _____.

- A. Algorithm
- B. Image batch
- C. Dataset cycle
- D. Epoch

Correct Answer: D

Explanation: An epoch is one full sweep through the training set. Models usually need multiple epochs to learn, but too many epochs on too little data leads to overfitting. The other options are unrelated terms - an algorithm is the learning method, not a training cycle.

Q4 [Type: MCQ | Difficulty: Medium | QID : 52745]

If a trained model classifies a new photo as "Cat" with 92% confidence, this means the model is guaranteed to be correct 92% of the time.

- A. True
- B. False

Correct Answer: B (False)

Explanation: Confidence reflects the model's internal certainty about its own prediction - not external accuracy. A model can be very confident and still wrong, especially when the new image differs from what it saw in training. High confidence is not a guarantee of correctness, which is one of the most important takeaways from working with image classifiers.

Q5 [Type: MCQ | Difficulty: Easy | QID : 52746]

When training your own model, it is important to evaluate on data the model has not seen during training - not just on the training data - because performance on training data alone can be misleading.

- A. True
- B. False

Correct Answer: A (True)

Explanation: A model can perform perfectly on training data without learning generalisable patterns. Evaluating on unseen data is how you check whether the model actually learned something useful or just memorised. This is why a held-out test set is essential - without it, you cannot tell memorisation from learning.

Q6 [Type: MCQ | Difficulty: Medium | QID : 52747]

When training your own model, more epochs (training cycles) always produce a better model - so you should train for as long as possible.

- A. True
- B. False

Correct Answer: B (False)

Explanation: Past a certain point, more epochs cause overfitting - the model starts memorising quirks in the training data that do not apply to new images. The session's Experiment 4 demonstrates this directly: with ~100 images and 50 epochs, test accuracy can drop compared to the same data trained for only 1 epoch. The right approach is to monitor validation performance and stop when it stops improving.

Q7 [Type: MCQ | Difficulty: Easy | QID : 52748]

What is Kaggle primarily used for in the ML world?

- A. A coding language used for building and deploying AI models in production
- B. A platform for sharing datasets, hosting competitions, and learning ML
- C. An online game where players train robots to complete various challenges
- D. A search engine for finding academic research papers on machine learning topics

Correct Answer: B

Explanation: Kaggle is the go-to source for public datasets, competitions, and tools used by AI enthusiasts, data scientists, and machine learning engineers. Today's Cat vs Dog dataset came from Kaggle. The other options confuse a platform with a language or describe unrelated services.

Q8 [Type: MCQ | Difficulty: Easy | QID : 52749]

Image classification models learn to distinguish categories by analysing visual clues such as:

- A. The filename and metadata tags attached to the image file
- B. The size of the image file measured in megabytes or kilobytes
- C. Shapes, colours, textures, and edges in the image
- D. The date and time the photo was originally taken by the camera

Correct Answer: C

Explanation: The session lists shapes, colours, textures, and edges as the visual features models actually use. The other options are metadata that the model does not see and that have nothing to do with what is in the photo.

Q9 [Type: MCQ | Difficulty: Medium | QID :52750]

The dataset folder structure separates training_set/cats/ from training_set/dogs/. Why does this folder structure matter?

- A.** It speeds up the computer's file system when loading large image datasets
- B.** Folder names serve as labels for the images they contain
- C.** It is required by all operating systems to store image files correctly
- D.** It prevents files from being accidentally deleted during the training process

Correct Answer: B

Explanation: Folder structure acts as the labelling system for supervised learning - the folder name is the label attached to every image inside it. The other options invent unrelated benefits that have nothing to do with ML.

Q10 [Type: MCQ | Difficulty: Medium | QID : 52751]

In Experiment 1 (1 image per class, 1 epoch), the model scores 100% on the two training images but only 1/4 on the test images. What is the most accurate description of what happened?

- A.** The model has truly learned what cats and dogs look like in general
- B.** The test images must be mislabelled or corrupted in some way
- C.** The model memorised the images rather than learning the actual concepts
- D.** The model needs to be tested on many more images before it can be evaluated fairly

Correct Answer: C

Explanation: With only one example per class, there are no patterns to learn - just two specific images to memorise. This is the textbook memorisation-without-learning case. Option A is the misconception being tested. Options B and D dodge the real issue: the dataset itself is too small to support learning.

Q11 [Type: MCQ | Difficulty: Medium | QID : 52752]

In the experiments, how did increasing the data from 10 images to around 100 images per class typically affect performance?

- A. The model's accuracy remained exactly the same regardless of the data amount
- B. The AI stopped working because it reached its maximum capacity at 50 images
- C. Performance decreased because the AI became confused by too many pictures
- Performance improved significantly as the model saw more varied examplesD.

Correct Answer: D

Explanation: A larger dataset provides more features and patterns for the AI to learn, leading to better generalisation. The other options invent fake limits or claim that more data hurts - both contradicted directly by the experiment results.

Q12 [Type: MCQ | Difficulty: Medium | QID : 52753]

Why is training data often split into training, validation, and test sets when building your own model?

- A. To save disk space on the computer running the training process and reduce the storage needed for large datasets
- B. To make the overall training process faster and more efficient by reducing the amount of data used
- C. Training teaches the model, validation tunes settings, test evaluates final performance on new data
- D. So the model can read and process multiple types of data simultaneously during training

Correct Answer: C

Explanation: The three-way split has distinct purposes: train (learn), validate (tune training settings), test (final unbiased evaluation). This separation is what prevents you from accidentally using your test set to make training decisions - which would inflate test accuracy without improving real performance.

Q13 [Type: MCQ | Difficulty: Medium | QID : 52754]

After Experiment 4 (~100 images, 50 epochs), a team observes that test accuracy dropped compared to Experiment 3 (~100 images, 1 epoch). The most likely explanation is:

- A.** The model finally became powerful enough to "see through" the test images and predict perfectly
- B.** Teachable Machine has a known bug that causes it to malfunction after 10 training epochs
- C.** The test images expired and are no longer valid for evaluating the model's performance
- D.** The model overfit - extra training cycles made it to memorise quirks that do not apply to new images

Correct Answer: D

Explanation: This is exactly what Experiment 4 warns about: more epochs on the same data can backfire by causing the model to memorise the training set rather than generalise. The other options are nonsense or fabrications.

Q14 [Type: MCQ | Difficulty: Medium | QID : 52755]

Why does the activity use the same 4 test images across all four experiments instead of different test images each time?

- A.** Keeping test images constant isolates the effect of changing data amount and epochs
- B.** Test images are expensive to produce, so reusing them across experiments saves time and resources
- C.** Teachable Machine only accepts the same test images repeatedly and cannot process new ones each round
- D.** Using new test images each round would be more accurate but takes far too much time to set up

Correct Answer: A

Explanation: A controlled experiment changes one variable at a time - that is why the tests stay fixed. If both the test images AND the training settings changed between experiments, you would not know whether the difference in results came from the training changes or from luck with different test images. The other options invent reasons that do not reflect experimental design.

Q15 [Type: MCQ | Difficulty: Hard | QID : 52756]

A student plots their model's accuracy across training: training accuracy keeps climbing (90% → 95% → 99%), but test accuracy peaks at epoch 12 and then decreases (85% → 87% → 82% → 75%). What pattern is this?

- A.** Underfitting - the model needs more training epochs to learn the patterns properly
- B.** Overfitting - past a certain point, more epochs hurt generalisation
- C.** Bias - the dataset is unfair and does not represent all categories equally
- D.** Garbage In, Garbage Out - the labels on the training data must be wrong or inconsistent

Correct Answer: B

Explanation: The signature overfitting curve: training accuracy keeps rising while test accuracy turns down. Underfitting would show low training AND test accuracy. Bias would show demographic skew, not a training/testing gap. GIGO would degrade training accuracy too, not just test accuracy.

Q16 [Type: MCQ | Difficulty: Hard | QID : 54141]

Two teams build cat-vs-dog models. Team A uses 200 images: 100 of one specific Persian cat and 100 of one specific Golden Retriever. Team B uses 200 images: 100 cats across many breeds, ages, and settings, and 100 similarly varied dogs. Both train for the same number of epochs. Which is most likely true on a varied test set?

- A.** Team A will outperform Team B because their model "knows" each specific image deeply and precisely
- B.** Both teams will perform equally well, since the total number of training images is identical
- C.** Team A will outperform Team B because Persians and Golden Retrievers are easier breeds to recognise
- D.** Team B will outperform Team A because their model learned what cats and dogs look like in general

Correct Answer: D

Explanation: Variety beats raw count when count is held constant. Team A's model only ever saw one Persian and one Golden Retriever, so it has no way to know that other breeds count as cats and dogs. Team B's model saw the same total number of images but with the variety needed to learn the general concept. Options A and C mistake depth-of-coverage-on-one-animal for general learning. Option B ignores the variety dimension entirely.

Q17 [Type: MCQ Scenario | Difficulty: Medium | QID : 54146]

Scenario: A team's model scores well on the test set but performs poorly when deployed in a real-world app where users upload phone photos of pets.

Question: What is the MOST likely cause?

- A.** The test_set images and real-world phone photos differ in ways the training data did not cover
- B.** The model needs many more training epochs to reach acceptable performance on real-world inputs
- C.** The model's underlying algorithm is fundamentally broken and cannot be used for image classification tasks
- D.** Teachable Machine cannot deploy models to real-world applications outside of its own platform

Correct Answer: A

Explanation: Passing a held-out test set does not guarantee performance on a different distribution - phone photos taken by random users look very different from curated dataset photos (different lighting, angles, quality, backgrounds). This is the deeper generalisation gap that the formal test set alone cannot catch. Option B is the lazy default. Options C and D blame the wrong components.

Q18 [Type: MCQ Scenario | Difficulty: Hard | QID :54152]

Scenario: A team's model achieves 99% accuracy on the test set, but on inspection they discover the model classifies any image containing grass as a "dog" - because most of their dog photos were taken in parks while most of their cat photos were taken indoors.

Question: Which statement BEST captures what went wrong?

- A.** The model needs to be trained for many more epochs to learn the difference between cats and dogs properly
- B.** The test set was too small and should be expanded to include more diverse cat and dog images
- C.** The model has overfit to a spurious feature (background) that happens to correlate with the label in this dataset
- D.** The model is biased against indoor photos in general and cannot recognise animals in indoor settings

Correct Answer: C

Explanation: A classic data-collection trap: the model learned the easiest discriminator (grass vs. indoor), which happened to be the background, not the animal. Option A would worsen the problem by reinforcing the wrong pattern. Option B misses the cause. Option D misnames the issue - the model is not "biased against" indoor photos in a fairness sense; it has just learned a useless shortcut feature.

Q19 [Type: FRQ - Short Answer | Difficulty: Hard | QID :54153]

Scenario: A high school team is training a model to identify invasive plant species from photos for their local nature preserve. They have 800 photos. After training, the model performs well on photos similar to the training set but struggles on photos from different seasons (when plants look different).

Question: Explain why this is happening and what the team should consider.

(Respond in 3–4 sentences.)

Model Answer: The model learned patterns from photos taken at specific times of year, but plants look different across seasons (different colours, leaf shapes, flowering states, growth stages), so when the model sees a plant in a different state than what was in training, it does not recognise it. The team should consider expanding their training data to include photos of the same species across different seasons and conditions, and should be transparent with the nature preserve about the model's limits - particularly that it may perform less reliably on photos taken in seasons not well-represented in their training data.

Q20 [Type: FRQ - Scenario Analysis | Difficulty: Hard | QID : 54155]

Scenario: A student wants to share their student-trained AI model online: a tool that predicts diabetes risk from a small set of inputs. The student trained the model on a small dataset they found online. They got 78% accuracy on validation. They're excited and want to share it as a free health tool. Their teacher suggests they think carefully before sharing.

Question: Identify one significant benefit AND one significant risk of sharing this kind of student-trained model publicly, and describe what the student should do - including whether you think this kind of student project should be shared publicly at all.

(Respond in 3–4 sentences.)

Model Answer: A significant benefit is that sharing student projects can demonstrate what is possible with accessible AI tools, contribute to public understanding of how machine learning works, and inspire other students to explore the field. A significant risk is that someone might take the tool at face value and use it to make decisions about their actual health - perhaps avoiding a doctor visit because the model said low risk, or worrying unnecessarily because it said high risk - when the model was trained on a small uncertain dataset, has not been validated by medical experts on real patient populations, and its 78% accuracy could mean very different things for different people, with errors affecting real health decisions. The student should not share this as a "health tool" but could share it as an educational demonstration of how ML works, with prominent and clear statements that the model is a student learning project, that it should NOT be used for health decisions, and that real medical AI requires extensive validation and clinical review - and should also consider whether the educational value is worth the risk that even with disclaimers, some users may rely on it inappropriately, particularly users without access to medical care otherwise.